

100 Short Questions with Answers - CCE 322

#	Question	Answer
1	The 8255A is a general purpose programmable I/O device designed to transfer the data from I/O to ____.	interrupt I/O
2	How many ports does 8255A have?	3 ports (PORT A, PORT B, and PORT C)
3	Which port can be split into two parts?	Port C (PC0-PC3 lower and PC7-PC4 upper)
4	What are the features of 8255A?	3x8-bit I/O ports (PA, PB, PC), TTL compatible, requires external demux of address/data bus
5	What is responsible for controlling the internal/external transfer of data/control/status word?	Read/Write Control Logic
6	Which of the following uses N-MOS technology? (8253, 8254, 8255, 8256)	8255
7	What is a tri-state, bi-directional, 8-bit buffer used to interface 8253/54 to system data bus?	Data Bus Buffer
8	In how many modes can 8253/54 be operated?	6 different modes
9	Which mode of 8253/54 can be used as a monostable multivibrator?	Mode 1: Programmable One Shot
10	Which mode of 8253/54 generates a strobe in response to an externally generated signal?	Mode 5: Hardware Triggered Mode
11	How many types of interfacing are there?	2 types (Memory interfacing and I/O interfacing)
12	In which communication, interface gets a single byte and sends it bit by bit serially?	Serial Communication Interface

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13	Which are known as Higher Address Bus?	A15-A8
14	In which mode does CPU periodically read internal flag of 8279 to check key press?	Polled mode
15	What is the function of Encoded Mode in 8279?	Counter provides binary count to be externally decoded for scan lines
16	Which pin is used to blank the display during digit switching?	BD (Blank Display)
17	Which mode allows 8/16 character multiplexed displays as dual 4-bit/single 8-bit?	Display Scan mode
18	What does DMA stand for?	Direct Memory Access
19	What are the features of 8257?	4 channels, 16-bit address and 14-bit counter, 64kb transfer per channel, independently programmable
20	What is the frequency range for 8257?	250Hz to 3MHz
21	An 8-bit microprocessor can process _____ data at a time.	8-bit
22	The number of bits processed in a single instruction is known as?	Bandwidth
23	Which is the pin efficient method of communication between devices?	Serial ports
24	The time taken for data transmission depends on?	Clock frequency and number of bits transferred
25	Which is the most commonly used buffer in serial porting?	FIFO buffer

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26	What does SPI stand for?	Serial Peripheral Interface
27	Which allows full duplex synchronous communication between master and slave?	Serial Peripheral Interface (SPI)
28	How many data lines does 256×4 have?	4 data lines
29	How is the number of chips required determined?	By number of data lines and width of data path from processor
30	Where is memory address stored in a C program?	In the pointer
31	Which term is used to refer to the order of bytes?	Endianness
32	What does SIMM stand for?	Single In-line Memory Module
33	Which can destroy the accuracy in algorithms?	Delays in memory management unit
34	Which step detects the key in a 4×4 keyboard matrix?	Mask bits individually, check if key pressed for >20 microseconds
35	In reading columns of a matrix, if no key is pressed, what do we get in binary?	All 1s (connected to power supply)
36	If a key operates in interrupt mode, what kind of interrupt does it generate?	External hardware interrupt
37	To identify which key is being pressed, what do we need to do?	Ground the pins one by one
38	How many data lines are there in a 16×2 alphanumeric LCD?	8 data lines (pin 7 to 14)

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39	Which pin of the LCD is used for adjusting its contrast?	Pin 3
40	For writing commands on an LCD, what is the RS bit?	RS pin is reset (0)
41	Which command of LCD is used to shift the entire display to the right?	0x1C
42	Which instruction is used to select the first row first column of an LCD?	0x80
43	Which pins of a microcontroller are directly connected with 8255?	RD, WR, D0-D7
44	Find control word for PA=out, PB=in, PCL=out, PCH=out (Mode0)	10000010b = 0x82H
45	Which pins are used to select ports and the control register?	CS, A0, and A1 pins
46	What is the value of control register when RESET button is set to zero?	Remains the same (all ports initialized as input when RESET is high)
47	Why is MOVX instruction used to access ports of 8255?	Because 8255 is in memory mapped I/O configuration, treated as external memory
48	What is BSR mode in 8255?	Bit Set/Reset mode used to program individual bits of PORT C only
49	How many pins of 8255 can be used as I/O ports?	24 pins (3 ports × 8 pins)
50	Resize 1024×768 to 640 pixels wide with same aspect ratio: height = ?	height = $640 \times 768 / 1024 = 480$

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51	What is an image's aspect ratio?	Ratio of its width to its height
52	With 2 bits per primary color in RGB, how many possible colors per pixel?	$2^2 \times 2^2 \times 2^2 = 64$
53	With 2-byte pixel values in a 24-bit lookup table, how many bytes for table?	$2^{16} \times 24/8 = 196,608$ bytes
54	What is Arduino?	Open-source hardware and software platform that is easy to use
55	How many types of Arduino do we have?	4 Arduino boards and 4 Arduino shields
56	What language is typical Arduino code based on?	C++ (lightly modified version)
57	What language is the Arduino IDE built on?	Java
58	How many analog pins are used in Arduino Mega board?	16 analog inputs
59	Arduino IDE consists of 2 functions. What are they?	setup() and loop()
60	Arduino codes are referred to as _____ in the Arduino IDE.	Sketches
61	What is the default bootloader of Arduino UNO?	Optiboot bootloader (512 bytes)
62	What is the use of Arduino.h header file?	Provides access to Arduino's features like GPIO, Analog IO, etc.
63	What is the use of Vin pin on Arduino Boards?	Accept 9V power supply or provide 9V output

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64	What is the execution process of Arduino code?	Editor → Preprocessor → Compiler → Assembly Instruction Set
65	What is the microcontroller used in Arduino UNO?	ATmega328p
66	What does 'p' refer to in ATmega328p?	Picopower technology
67	Which board is the first to use a microcontroller with built-in USB?	Arduino Leonardo
68	Which Arduino Board contains an onboard joystick?	Arduino Esplora
69	What is the function of IOREF pin on Arduino UNO?	Provides operating voltage reference
70	Which processor supports Arduino Zero?	Atmel Smart SAM D21 (32-bit ARM Cortex M0+)
71	Which software is used to upload Arduino Sketches?	avrdude software
72	What is the use of 2 serial pins on Arduino Diecimila?	TX (transmission) and RX (receiving) for Serial TTL Signals
73	Which Arduino Boards use Atmega2560?	Arduino Mega and Arduino Mega ADK
74	What is the operating voltage of Atmega328?	1.8V to 5.5V
75	Which Arduino Boards use Atmega32U4?	Arduino Leonardo and Arduino Pro Micro
76	Is Arduino code Object-Oriented or Procedural?	Object-Oriented (subset of C++)
77	What is the difference between IDE and compiler?	Compiler: debugging and executing code; IDE: environment for writing code

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78	What will be the output when modifying a #define value?	Compiler will throw an error
79	Where does Arduino IDE search for board name information?	boards.txt file
80	How many times does setup() function run on startup?	Once
81	If pin2 receives "1011", what is output of pin1?	Inverse of the wave: "0100"
82	Which Arduino Board does SigFox Library work with?	Arduino MKRFOX1200 Board
83	Output of sizeof(a + d) where a=int, d=double?	8 bytes
84	What is the use of ESP8266 WiFi Module?	Connects microcontroller to WiFi network
85	Why is Arduino Mega more viable for complex projects than UNO?	More I/O pins (54 vs 14), more PWM (14 vs 6), more memory (256KB vs 32KB Flash, 4KB vs 2KB SRAM)
86	Atmega168 is an ____ bit chip.	8-bit
87	Why does every compilation check for previous compilations?	To use previously generated ".o" files and reduce compilation time
88	What type of signal does analogWrite() function output?	Pulse Width Modulated (PWM) signal
89	What is the objective of code using EEPROM.write()?	Fills EEPROM with 1's and turns on onboard LED
90	What is the use of SD.h Library?	Allows communication between Arduino and SD Card Module

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91	What is the use of Ethernet library?	Connects Arduino to internet via Ethernet connection
92	Purpose of Serial.write(40) code?	Sends data in form of a byte with value 40 using UART
93	Which chipset is the LCD library for Arduino based on?	Hitachi HD44780 Chipset
94	What is the resolution of micros() function on Arduino Nano?	Depends on frequency (4 microseconds on Duemilanove)
95	What is the use of Interrupt Service Routine in Arduino?	Pauses current function and diverts memory to another function
96	Output when 5V supplied to A0 pin on Arduino UNO?	1024
97	What mode should Arduino pin be in for ultrasonic sensor object detection?	Digital mode (TRIGGER HIGH for 10μs, ECHO reads digital)
98	What happens if 25V is supplied to Nokia 5110 GDM Vcc?	May suffer permanent damage (works on 3.3V-5V)
99	Which port should connect to Key Port on Adafruit FONA 3G+GPS?	Gnd pin
100	How to throw an error using preprocessing directives?	Using #error directive